How Foveated Vision Shapes Cognitive Maps in Navigation Agents

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Paper under double-blind review

Abstract

The same neural circuitry codes different things in animals that see differently: primate hippocampus encodes joint place–view–action state because primates sample the world through a foveated sensor; rodents with near-panoramic vision develop classic place fields; bats remap under vision vs. echolocation. The biological pattern is that *sensor structure shapes the format of spatial memory*. Whether the same is true of artificial navigation agents has not been tested directly: prior work has either removed vision or used spatially uniform input. Holding task and architecture fixed and varying only the visual sensor across a five-condition spectrum (no vision, a resolution-collapsed encoder, foveated vision with fixed and learned gaze, and full uniform vision), we ask whether sensor structure functions as a representational-content axis on learned spatial memory. We report evidence that it does. An *encoder–memory race* emerges: when the visual encoder supplies little spatial-feature variety per step, the recurrent state compensates with a linearly-decodable world-frame position code; when the encoder is rich, it substitutes — the integrated code emerges transiently in early training and progressively loses linear top-layer readability as the visual route consolidates. Agents at comparable task competence develop hidden states in linearly disjoint subspaces, and cross-condition memory transplants are asymmetric: bottleneck-donor states hurt rich-encoder recipients more than vice versa. The intuition that "richer perception begets richer memory" fails in this regime, suggesting encoder capacity as a candidate training-time lever for cognitive-map-like representations. The pattern echoes biological findings on hippocampal compensation and cross-modality remapping — convergent (not conclusive) evidence that sensor–memory coupling is a principle worth taking outside this single architecture.

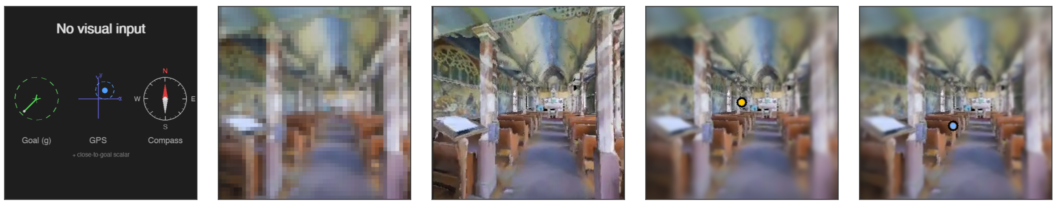
1 Introduction

Primate spatial memory is shaped by how primates see. Hippocampal cells code joint place–view–action state in part because the primate eye samples the scene through a foveated sensor [Wirth et al., 2017, Rolls, 2024]; rodents with near-panoramic vision instead show classic place fields. Sensor structure also reorganises memory *within* a single animal — bats produce non-overlapping hippocampal populations under vision vs. echolocation [Geva-Sagiv et al., 2016], and congenitally blind humans develop enhanced hippocampal coupling on non-visual spatial tasks [Kupers and Ptito, 2014]. The biological pattern is consistent: *the structure of the sensor reshapes the structure of spatial memory*. We ask whether the same is true of artificial navigation agents.

The emergent-cognitive-maps literature has shown that map-like structure arises in the learned internal state of navigation agents trained with deep RL, including agents with no vision at all [Wijmans et al., 2023, Banino et al., 2018, Cueva and Wei, 2018].[1] These results establish that *some* map-like structure

[1] "Cognitive map" here means a linearly-decodable world-frame position code in the hidden state — the operational quantity our probes measure. The broader-literature concept [Epstein et al., 2017] (relational,

is learned without being designed in. They leave open the complementary question we take up here: holding task and architecture fixed, does the structure of the visual sensor reshape *what* the agent's memory encodes? We isolate sensor structure as the only varied component across five conditions spanning a sensor-structure spectrum (Figure 1): no vision, a resolution-collapsed encoder, foveated vision with fixed gaze, foveated vision with learned gaze, and uniform full-resolution vision. The resolution-collapsed control distinguishes outright blindness from the case where vision is present but the encoder cannot spatially resolve it; foveation occupies a biologically motivated point in between. *We use a recurrent (LSTM-based) PointNav agent as our case study*: this matches the architecture of the emergent-maps literature we extend [Wijmans et al., 2023], gives every cross-condition comparison a single hidden-state object to probe, and provides a tractable controlled setting. The principle we test — whether sensor structure shapes the format of learned memory — is stated in architecture-agnostic terms; whether it transfers to transformer-based or supervised visual learners is a scope question we discuss in §5, not establish here.



(a) Blind (b) Coarse (1×1) (c) Uniform (d) Foveated (fix) (e) Foveated (learned)

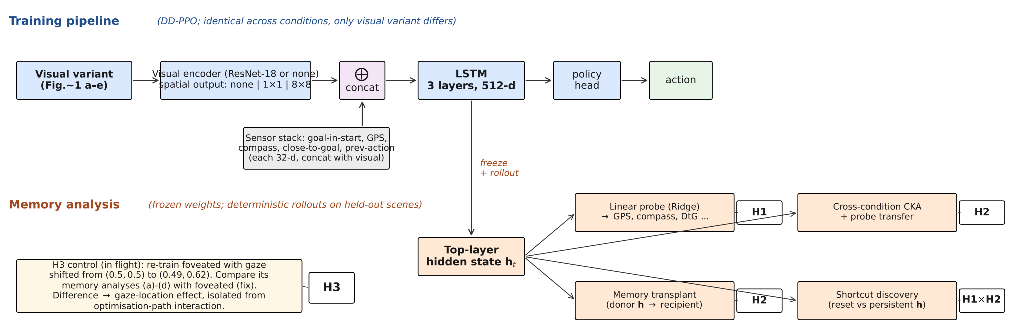


Figure 1: **Setup.** *Top row, (a)–(e):* the five visual sensor conditions, shown on the same scene. (a) *Blind*: no visual stream; only the non-visual sensor stack. (b) *Coarse* (1×1): 48×48 uniform RGB, chosen so the ResNet-18 encoder collapses spatially to a 1×1 feature map. (c) *Uniform*: 256×256 full-resolution RGB. (d) *Foveated (fix)*: eccentricity-dependent Gaussian blur ($\sigma_{\max} = 8$, quadratic falloff) centred at the image middle (yellow dot). (e) *Foveated (learned)*: same blur model with a learned (x, y) gaze offset (blue dot, post-training collapsed location $(0.49, 0.62)$). *Bottom block:* the structural variable being manipulated is encoder spatial output (none | 1×1 | 8×8); the DD-PPO learner, sensor stack, LSTM backbone, and policy head are otherwise identical. After training, weights are frozen and we collect deterministic rollouts to extract top-layer hidden states $\mathbf{h}_t$, on which we run linear probes (H1 evidence), cross-condition CKA + probe transfer (H2 evidence), memory transplants (H2 behavioural validation), and shortcut discovery (probe-readable vs. policy-used dissociation, §4.5).

We test three hypotheses. *H1 (encoder–memory race)*: when the visual encoder supplies little spatial-feature variety per step, the recurrent state should compensate by carrying a world-frame spatial code; richer encoder output should let memory rely on visual features instead. *H2 (format divergence)*: different sensor conditions should produce hidden-state representations in distinct linear subspaces, not scaled versions of a common code. *H3 (gaze location)*: within the rich-encoder regime, where a foveated sensor is centred should act as a second content axis independent of the sensor transform. Each has a direct biological analog (§5.2).

hierarchical maps) is wider; richer non-linear structure may exist in the same hidden states but is outside our linear-probe scope.

Contribution. We propose that *sensor structure* acts as a representational-content axis on the spatial memory learned by recurrent navigation agents, and that this axis has at least three orthogonal dimensions: a magnitude axis (encoder-bandwidth-driven memory recruitment, H1), a format axis (linearly disjoint condition-specific subspaces, H2), and a gaze-location axis (where attention points within the rich-encoder regime, H3). The claim is stated in interface-level terms (upstream-encoder $\rightarrow$ downstream-memory), isolated in a controlled five-condition silicon case study, and discussed alongside convergent biological evidence (§5.2).

Three findings emerge with the strongest evidence. An encoder–memory race along the sensor-structure spectrum (H1) that lives inside the recurrent state, not at the encoder; cross-training probing confirms the race plays out as a substitution effect within training (rich-encoder agents transiently learn an integrated spatial code, then unlearn its linear top-layer readability as the visual route consolidates). The five agents solve the task at comparable competence but develop hidden states in linearly disjoint subspaces, with measurable cross-condition memory-transplant cost (H2). And probe-readable vs. policy-used memory dissociates in opposite directions for two specific conditions, suggesting that "has a cognitive map" and "uses a cognitive map" need not coincide. Foveation under our blur model converges with uniform on H1 but diverges on H2, behaviour, and dataset shift. H3 (gaze location) is tested by a foveated-shifted control and a stochastic-gaze variant whose results appear in §4.6.

The cross-condition pattern is supplemented by a causal H1 test — a scaling-sweep experiment varying the encoder's input resolution from 32^2 to 192^2 pixels (Appendix H) — and by foveation-specific predictions involving alternative foveation models (§4.4, Appendix G). Each finding has a direct biological precedent which we discuss as convergent (not conclusive) evidence in §5.2.

2 Related Work

Cognitive maps in RL navigation agents Wijmans et al. [Wijmans et al., 2023] demonstrated that map-like representations emerge in the recurrent memories of DD-PPO agents even without visual input; Dwivedi et al. [Dwivedi et al., 2022] probed sighted agents for related spatial concepts; Banino et al. [Banino et al., 2018] and Cueva & Wei [Cueva and Wei, 2018] found grid-cell-like units in recurrent networks trained on path integration; Gornet & Thomson [Gornet and Thomson, 2024] extended this via visual predictive coding. Hennig et al. [Hennig et al., 2023] showed that RL under partial observability can induce belief-state-like representations without explicit probabilistic objectives. Ramakrishnan et al. [Ramakrishnan et al., 2025] examined spatial cognition in frontier models. All these studies use either absent or spatially uniform perception; none vary input quality spatially, and none test whether different sensors produce different representational formats.

Foveated vision and gaze policies in deep learning Foveated rendering [Strasburger et al., 2011, Rosenholtz, 2016] exploits eccentricity-dependent acuity falloff for efficiency. Learned glimpse policies appear in recurrent attention models [Mnih et al., 2014]; Atanov et al. [Atanov et al., 2024] showed that sensor layout changes downstream task performance. Pourrahimi & Bashivan [Pourrahimi and Bashivan, 2025] probed a foveated visual-search model for neuroscience-predicted features; Shang & Ryoo [Shang and Ryoo, 2023] jointly learned gaze and motor policies following the Sprague & Ballard [Sprague and Ballard, 2003] framework. These address visual search or task performance; none asks how sensor structure reshapes the learned spatial memory of a downstream navigation policy.

Probing neural representations Belinkov [Belinkov, 2022] surveys probing classifiers; Hewitt & Liang [Hewitt and Liang, 2019] introduced control tasks to distinguish genuine structure from probe memorisation; Kornblith et al. [Kornblith et al., 2019] introduced linear CKA for cross-model similarity. We adopt linear probing, the Hewitt–Liang selectivity metric, and unaligned linear CKA, and complement them with two behavioural interventions directly on the recurrent state (memory transplant and shortcut discovery) to validate the probe-based findings outside the linear-probe framework.

What is new here Three things distinguish this work. (i) A *controlled* five-condition ablation with everything but the visual pathway held identical, including the coarse condition that decouples "no vision" from "vision the encoder cannot resolve". (ii) A protocol-corrected probing methodology (deterministic action selection at probe time; stochastic sampling inflates probe R^2 via a 4-step quasi-static-trajectory artefact, Appendix A). (iii) Convergent behavioural validation via memory transplant and shortcut discovery, outside the linear-probe framework. Biological motivation and the parallel to animal-navigation research are integrated in §5.2 rather than recapitulated here.

3 Methods

3.1 Setup

All agents are trained on PointGoal navigation in Habitat [Savva et al., 2019] using DD-PPO [Wijmans et al., 2020] on a merged pool of Gibson-0+ [Xia et al., 2018] (411 scenes) and Matterport3D train (61 scenes), totalling 472 training scenes; evaluation uses 18 held-out MP3D test scenes (1008 episodes). All agents share an identical 3-layer LSTM (512-d) backbone and the Wijmans non-visual sensor stack (goal-in-start-frame, GPS, compass, close-to-goal scalar), each projected to 32-d and concatenated with a 32-d previous-action embedding. *Frame budgets vary by per-condition convergence rate, not by experimental design:* sighted conditions converge in $\sim$250M environment frames; blind requires $\sim$342M (no visual input $\rightarrow$ slower memory-integration convergence; the same convention is followed by [Wijmans et al., 2023]); foveated (fix) uses ckpt.36 at $\sim$174M as its last clean intermediate (a silent NaN-gradient corruption affected later checkpoints; Appendix F). All cross-condition comparisons in §4 are made *at convergence* (success $\in [0.93, 0.99]$ across all five conditions), not at frame-matched checkpoints; the H1/H2 differences we report are at-converged-behaviour representational differences, not undertraining differences.[2]

The five visual conditions (Figure 1): *blind* (no visual input); *uniform* (full 256×256 RGB encoded by ResNet-18); *foveated (fix)* (eccentricity-dependent Gaussian blur, $\sigma_{\max} = 8$ quadratic falloff, gaze locked at image centre, same ResNet-18 encoder)[3]; *coarse* (48×48 uniform RGB; at this resolution ResNet-18's 32-fold spatial downsampling collapses the feature map to 1×1, removing spatial structure but preserving channel information); *foveated (learned)* (same blur model, gaze centre predicted by a lightweight MLP trained end-to-end). Coarse is the spatial-bandwidth lower bound for our encoder; it has visual input but its encoder output cannot resolve world-frame position. The 48×48 input also has lower spatial resolution than uniform's 256×256, so this single condition cannot fully separate the 1×1 collapse from low input resolution; the encoder-resolution scaling sweep (Appendix H) addresses this directly. The foveated conditions disable the running-mean-and-variance input normaliser to avoid an in-place autograd conflict along the gaze-decoder gradient path; an invariance control is in training. We also discovered and patched a silent NaN-gradient corruption mode in DD-PPO (Appendix F).

3.2 Probes

After training, all weights are frozen. We collect per-step top-layer LSTM hidden states ($\mathbf{h}_2$, 512-d) paired with ground-truth labels from n=500 Gibson held-out episodes under *deterministic* argmax-action rollouts (matching our shortcut / transplant / standard eval protocols; the alternative stochastic protocol gives trivial ~ 4-step rollouts and is documented in Appendix A).[4] Ridge probes ($\alpha = 10$) are fit with episode-level 5-fold cross-validation; we report mean $\pm\sigma$ across folds and verify probe specificity with Hewitt–Liang label-permutation selectivity [Hewitt and Liang, 2019].

[2] Algorithm-and-task scope: DD-PPO + PointGoal in Habitat with ResNet-18 visual encoder. The H1/H2 patterns we report could in principle shift under different RL algorithms (A2C, IMPALA, SAC), different navigation tasks (ObjectGoal, ImageGoal, exploration), different simulators (iGibson, AI2-THOR), or different visual encoders. Each is a deliberate scope choice for direct comparability with [Wijmans et al., 2023]; generality is open.

[3] Foveation-model choice: Gaussian blur with quadratic eccentricity falloff is a tractable approximation, not a faithful model of biological retinal sampling. Real foveation uses spatially non-uniform sampling (log-polar or multi-scale pyramid) which directly reduces the encoder's spatial output dimensionality, whereas Gaussian blur preserves it. The F3 log-polar control (Appendix G) is a falsifiable test of whether the encoder spatial-output story explains foveation under a different operationalisation; if log-polar matches uniform on H1, the mechanism story would need reframing.

[4] Main-paper results probe $\mathbf{h}_2$ specifically because it is what the linear policy head reads. Appendix Figure 10 reports Ridge probes across all six (hidden, cell) $\times$ (L0, L1, L2) representations: cell states $\mathbf{c}_t$ do not carry the GPS code as cleanly as hidden states, so the H1 "linearly-decodable top-layer GPS" claim is about $\mathbf{h}_2$ specifically. The full CKA / transplant / shortcut machinery on $\mathbf{c}_t$ and $\mathbf{h}_0, \mathbf{h}_1$ is left for follow-up.

The probed variables map to the three results sections. *H1 evidence (§4.2):* GPS, compass, distance-to-goal, lag-k path history ($k \in \{0, \dots, 20\}$). *H2 evidence (§4.3):* linear CKA [Kornblith et al., 2019] between conditions, cross-condition probe transfer (train probe on condition A, test on B), and 1-NN purity on pooled hidden states. *Boundary / null probes (§4.7):* visited-region occupancy, metric 20×20 occupancy grid, ego-frame goal vector, per-unit spatial information.

We complement probes with two interventions on the LSTM state. *Memory transplant*: paste a donor's hidden state into the recipient at step 30 and compare baseline / self-transplant / cross-transplant SPL across 150 episodes per pair — the cross-vs-self gap isolates condition-mismatch from rollout-divergence noise.[5] *Shortcut discovery*: run paired same-scene-different-goal episodes (20 scenes $\times$ 10 pairs) and compare second-episode SPL under reset vs. persistent LSTM — the SPL drop indexes how tightly the policy couples actions to integrated recurrent memory (used in §4.5 to compare "probe-readable" vs. "policy-relied-on"). For foveated (fix), all analyses use ckpt.36 at $\sim$174M frames (last clean intermediate before the silent-corruption window).

4 Results

4.1 Five conditions, same task

Table 1 collects the headline per-condition numbers. *All five agents solve the same task at near-equivalent competence.* Sighted conditions reach 96–99% success; blind reaches 93% at a longer training budget, reproducing Wijmans et al. [Wijmans et al., 2023]. The narrow success bracket rules out a skill-gap account for any encoded-content differences below. The probe machinery is calibrated: distance-to-goal decodes everywhere ($R^2 \in [0.81, 0.90]$), demonstrating Ridge has the capacity to fit high-R^2 scalars when the hidden state contains them; Hewitt–Liang label-permutation selectivity is high precisely where GPS is high (Appendix A).A classical Bug-style controller (greedy goal-seeker with always-right wall-recovery, same GPS+compass sensor stack, no learned weights) achieves SPL XX.X / success YY.Y% on the same Gibson eval scenes (Wijmans Table 1 reports an analogous Clairvoyant-Bug at 46% SPL on their setup); contextualises the 59–85% SPL range our agents reach. Result in flight. The behavioural near-equivalence sets up the question the rest of §4 addresses: *what differs internally*, given that what the agents *do* is the same?

Table 1: Per-condition headline. Probe R^2 columns are the 5-fold episode-level CV mean $\pm$ std on full-length deterministic-rollout trajectories (no per-episode step cap). *GPS* and *Compass* are world-frame. The bottleneck-vs-rich-encoder gap is in temporal *stability* of the code, not its mid-episode existence (Figure 2b): all five conditions decode top-layer GPS at $R^2 \in [0.6, 0.95]$ in the typical-episode window (steps 50–200); the rich-encoder long-tail collapse is what produces the sub-chance numbers below. Bold marks the H1-relevant ordering (blind > coarse $\gg$ rich-encoder).

Condition	Frames	SPL	Success	GPS R^2 (no-cap)	Compass R^2 (no-cap)
Blind	342M	0.59	0.93	**+0.95±0.02**	**+0.81±0.08**
Coarse	250M	0.67	0.98	**+0.78±0.10**	**+0.64±0.10**
Uniform	250M	**0.85**	**0.99**	−0.31±0.86	+0.36±0.23
Foveated (fix)	174M	0.78	0.96	+0.06±0.88	+0.07±0.69
Foveated (learned)	250M	0.82	0.98	−2.43±3.98	−1.34±3.14

4.2 Encoder–memory race: a temporal-stability claim (H1)

The headline finding is non-obvious: visual-encoder capacity tracks top-layer LSTM spatial encoding *inversely*. Conditions with the smallest visual-feature throughput per step (blind, coarse) carry a strong linear GPS code at the policy readout, and conditions with the richest visual stream (uniform,

[5] Transplant midpoint t=30 is a single design choice. The supplementary midpoint sweep across $\{0, 15, 30, 60, 90, 200, 400, 800\}$ steps (Figure 9) shows the cross-self gap peaks in $t \in [30, 90]$ (mismatch matters most when the agent has integrated some history but is not yet committed) and decays toward 0 at $t \geq 400$. Earlier midpoints ($t < 15$) sit at a protocol-noise floor.

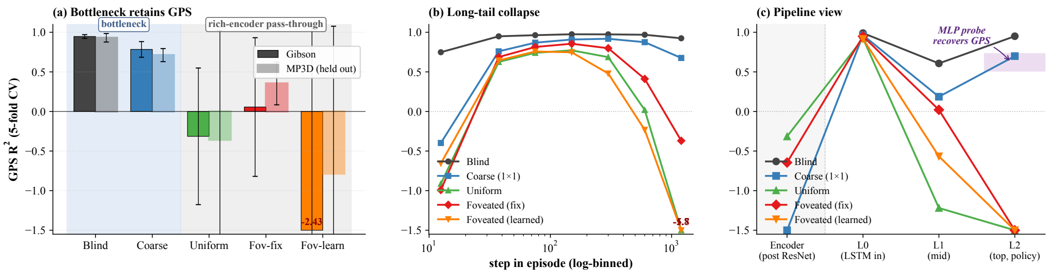


Figure 2: H1 evidence in three panels. *(a)* Top-layer GPS R^2 (5-fold CV, deterministic rollouts) on Gibson and held-out MP3D (300 episodes/condition; lighter bars = MP3D). Blind and coarse (left, blue background) decode world-frame GPS on both datasets; uniform / foveated / foveated (learned) (right, grey background) do not, except foveated recovering on MP3D. Negative-R^2 values that fall outside the y-axis range are annotated above the bar (e.g. Foveated (learned) at -2.43). *(b)* Top-layer GPS R^2 binned by step-in-episode: 7 logarithmically-spaced bins in $[10, 1600+]$ steps, with bin midpoints used for x-coordinates; the right-most bin pools all steps ≥ 1024. All five conditions encode top-layer GPS in the typical window (steps ~ 25–200); the code becomes no longer linearly readable in the rich-encoder long-tail (negative R^2 in the $800+$ step bin; clipped values labelled), while bottleneck conditions hold $R^2 > 0.7$ to step $800+$. The headline in (a) is therefore primarily long-tail behaviour. *(c)* GPS R^2 along the agent's information pipeline: *Encoder* (post-ResNet-18 feature-map probe, pre-LSTM) $\rightarrow$ *L0* (LSTM input layer, where the sensor stack concatenates with the visual encoder output) $\rightarrow$ *L1* (middle) $\rightarrow$ *L2* (top, policy-readable). For sighted conditions, the encoder is at chance/negative (no sighted encoder linearly carries GPS); the L0 jump is the GPS-sensor concat. The H1 divergence is at L2: bottleneck (blind, coarse) hold; rich-encoder discard. The shaded "MLP probe recovers GPS" band at L2 ($R^2 \in [0.51, 0.73]$) shows that a 2-layer MLP *can* decode GPS from rich-encoder L2 — the linear-probe failure is precisely about *linear* top-layer readability, not absence of GPS information. Blind has no encoder; foveated (learned) lacks the encoder probe but follows the L0–L2 pattern.

foveated, foveated (learned)) do not (Figure 2a). Goal-distance, included as a sanity check, decodes everywhere.[6]

Temporal stability is the precise claim Step-binned probing (Figure 2b) separates "never had a top-layer GPS code" from "had one mid-episode then lost linear readability"; rich-encoder conditions are the latter. The Table 1 "rich-encoder $\approx$ chance" headline is therefore long-tail behaviour, not mid-episode absence. *H1 is most precisely a temporal-stability claim about the linearly-decodable top-layer GPS code.* A lag-k probe corroborates: blind sustains GPS at $k = 20$, while rich-encoder conditions are at chance for every $k \leq 10$.

Pipeline view: where GPS sits, and where it disappears. Panel (c) of Figure 2 traces GPS R^2 across four locations along the agent's information path. Three readings come out together. *(i) The visual encoder does not linearly carry GPS*: encoder feature-map probes for coarse, uniform, and foveated all sit at or below chance ($R^2 = -3.14, -0.32, -0.65$ respectively). The H1 divergence is therefore not an encoder-level position-retention property — the encoders are roughly equally GPS-blind. *(ii) The L0 GPS comes from the sensor concat*, not the network "computing" position: every condition sees a sharp jump from encoder to L0 ($R^2 \rightarrow\sim 0.95$) because the GPS sensor embedding enters the LSTM input vector at L0. *(iii) The H1 divergence is specifically at the policy readout (L2)*: bottleneck conditions hold their L0 GPS through L2; rich-encoder conditions discard it. Since the DD-PPO action head is a linear projection of $\mathbf{h}_2$, this maps directly to "policy-accessible". A 2-layer MLP on the same L2 states recovers GPS for rich-encoder conditions ($R^2 \in [0.51, 0.73]$), so H1 is precisely about *linear* L2 decodability.[7] The blind result ($R^2 = 0.95$) replicates Wijmans et al. [Wijmans et al.,

[6] Whether the correlation reflects causation is what the §4.4 experiments and the Appendix H resolution sweep test directly. All numerical claims in §4.2 are single-seed; multi-seed replication of all five conditions is in flight.

[7] Two single-seed cross-dataset shifts on MP3D — foveated GPS at chance $\rightarrow +0.35$, foveated (learned) compass $-1.34 \rightarrow +0.41$ — await multi-seed replication; if they survive, they would suggest condition-specific MP3D recovery worth investigating, but at $N = 1$ we cannot rule out per-seed noise. The MLP result alone cannot exclude exploitation of position-correlated features; Appendix B discusses both.

2023]; coarse extends their finding to a condition that has visual input but a 1×1 encoder output, so the shared trigger is minimal visual-feature variety per step, not absence of vision.

Substitution mechanism, with cross-training evidence H1 is a substitution effect: integrating the Layer-0 GPS-sensor input across time and exploiting the encoder's position-correlated visual features are alternative routes to the same policy output. Bottleneck conditions force the integration route (no rich visual features available), so the temporally-stable GPS code persists; rich-encoder conditions provide a viable visual route, so the integrated code is no longer needed by the policy and ceases to be linearly readable. Probing each condition at 4–5 training checkpoints (Figure 3; 500 episodes per probe, 5-fold episode-level CV) supports this account at the level of the cross-condition pattern. *All three rich-encoder conditions* show a clear early-positive GPS R^2 (uniform $+0.79 \pm 0.07$, foveated $+0.79 \pm 0.09$, foveated (learned) $+0.67 \pm 0.18$ at $\sim$ 50M frames; tight 5-fold spread $\sigma \leq 0.18$, comparable to bottleneck conditions' early-training spread, indicates a real signal rather than a sampling artefact) followed by a regime where the same probe gives *no longer reliably positive* R^2 with much wider spread ($\sigma \in [0.4, 4.2]$ at later ckpts), consistent with probe extrapolation failure when no linear signal is present. Bottleneck conditions stay tight and high throughout: blind holds $R^2 \approx 0.95$ ($\sigma \leq 0.02$) across 50–340M frames, coarse $R^2 \in [0.68, 0.88]$ ($\sigma \leq 0.19$) across 50–250M frames. The three rich-encoder conditions also differ in *decay rate*: uniform's GPS R^2 has already left tight-positive by $\sim$ 100M frames, foveated still has $R^2 = +0.78$ at $\sim$ 100M frames before crashing by $\sim$ 150M, and foveated (learned) decays in between. We read this as the visual-route-substitution timescale tracking encoder informativeness about position — uniform substitutes fastest, foveated slower because peripheral acuity is degraded. The early-then-decay contrast is therefore both *within-episode* (Figure 2b) and *across training*: rich-encoder agents do learn an integration code transiently, then unlearn its linear top-layer readability as the visual route consolidates.[8]

Memory-length budget sweep (Wijmans 2023 Fig 2 replication). If the substitution-mechanism story is correct, bottleneck conditions need long-horizon LSTM memory (no rich visual route to substitute), while rich-encoder conditions can derive top-layer features from short history. We test this directly by clipping the LSTM hidden state to zero every K within-episode steps for $K \in \{1, 4, 16, 64, 256, 1000\}$ and re-probing GPS R^2 on the resulting K-truncated states (Appendix Figure **??**, single seed, 100 episodes per condition$\times K$). The expected signature: Blind / Coarse R^2 grows monotonically with K and saturates only at $K \sim 100+$ steps (consistent with Wijmans's Figure 2: blind navigation does not saturate until $K \sim 1000$); Uniform / Foveated R^2 stays low at every K because the encoder substitutes for memory integration. Numerical results forthcoming when in-flight cluster jobs land.

4.3 Format-level divergence across conditions (H2)

Behavioural test (memory transplant) Pasting a donor's mid-episode hidden state into a recipient's policy at step 30 tests whether the representational geometry differences *matter* outside the probe framework (Figure 4, right). Three structural facts emerge that no linear-similarity measure on its own would surface. *Asymmetry along the bottleneck axis:* bottleneck-donor states are toxic to rich-encoder recipients while the reverse direction is benign — the rich-encoder LSTM cannot make sense of bottleneck-donor activity at the policy readout, but a bottleneck LSTM can still drive its policy from rich-encoder donor activity. *Recipient brittleness scales with encoder bandwidth:* uniform suffers the largest cost from *any* foreign donor; blind tolerates them all. *The linear-similarity blind spot:* pairs that read as indistinguishable under unaligned linear CKA differ by 0.01–0.38 SPL under transplant — "these representations look identical to a linear similarity metric" is consistent with "their actual policy-relevant content is incompatible".[9]

Convergent evidence at the probe level. Cross-condition probe transfer (Figure 4, left) shows the same asymmetric pattern: probes trained on bottleneck states predict off-manifold by a wider margin on rich-encoder hidden states than the reverse. The 1-NN purity is 1.000 vs. chance 0.20 on 1500×5

[8] Per-checkpoint probes are single-seed; multi-seed replication is in flight. The decay-rate ordering is the part most exposed to seed variability. A direct causal test — ablating the visual stream mid-rollout in rich-encoder agents to see whether the top-layer GPS code re-emerges, or perturbing the GPS sensor input mid-rollout in bottleneck agents to see whether SPL collapses — is left for future work.

[9] Single-seed; the asymmetry pattern is qualitatively robust across the full 5×5 matrix but specific cell magnitudes will move with replication.

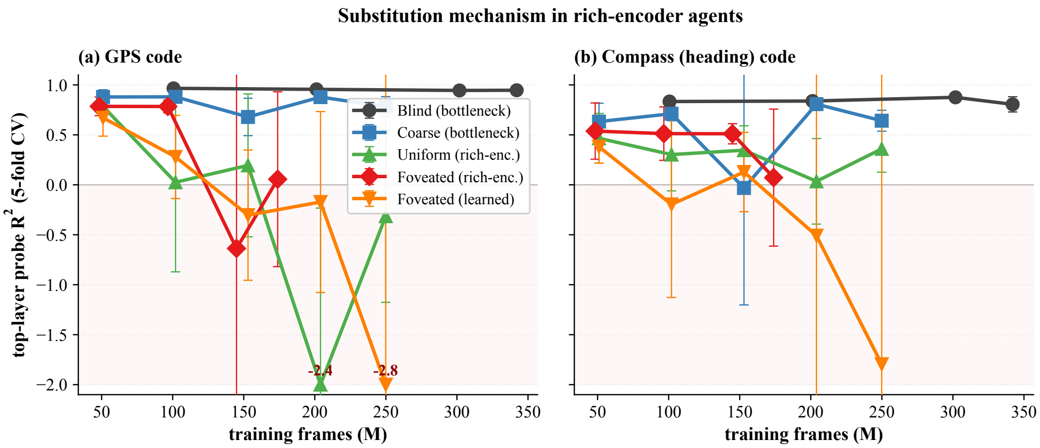


Figure 3: **Substitution mechanism: cross-training profile, GPS and compass codes.** Top-layer probe R^2 at 4–5 training checkpoints per condition (500 episodes per probe, 5-fold episode-level CV; single seed; partial-data ckpts shown with hollow dotted markers). *(a) GPS code:* bottleneck conditions (blind, coarse) hold tight, stably high R^2 across the full training trajectory; rich-encoder conditions (uniform, foveated, foveated (learned)) all start positive and tight at $\sim$ 50M frames ($R^2 \in [0.67, 0.79]$, $\sigma \leq 0.18$), then transition into a regime where R^2 floats around zero with much wider 5-fold spread, consistent with probe extrapolation failure when no linear signal is present. *(b) Compass code:* same qualitative pattern (bottleneck stable, rich-encoder dissipates), but with two informative differences — (i) uniform's compass code partially survives ($R^2 \approx +0.36$ at convergence) where its GPS code does not, suggesting heading is recoverable from the visual route while position is not; (ii) foveated (learned)'s compass collapses to deeply negative, paralleling its GPS. The substitution affects spatial codes broadly, not GPS specifically. Negative-R^2 outliers clipped at -2.0 for visibility.

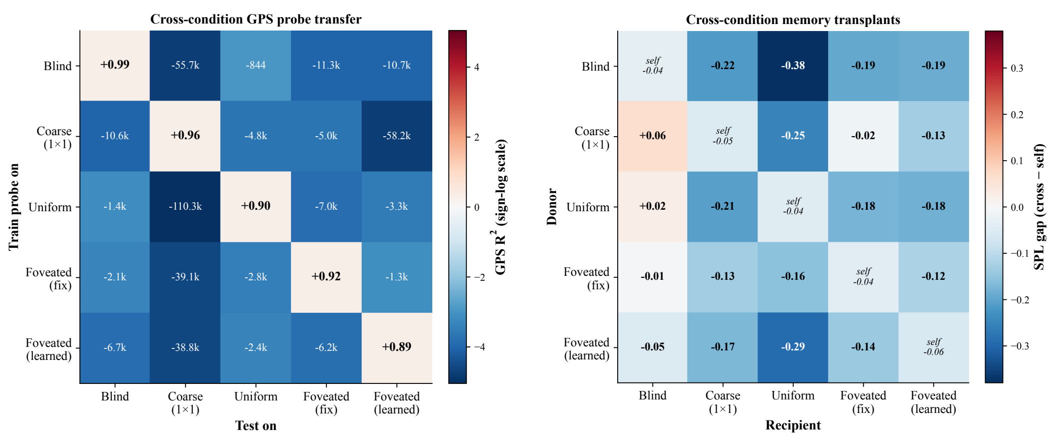


Figure 4: H2 convergent evidence, two views. *Left:* cross-condition probe transfer for GPS R^2 (train probe on row, test on column; signed-log colour scale). The diagonal is $+0.89$–$+0.99$ (self-probe), every off-diagonal predicts off-manifold ($R^2 \ll -800$, often $\ll -10^4$). *Right:* full 5×5 cross-condition memory-transplant matrix at midpoint t=30 steps; off-diagonal cells = cross-transplant SPL $-$ recipient's self-transplant SPL (isolated condition-mismatch effect). All 20 off-diagonal cells populated (single-seed). Both heatmaps are asymmetric: bottleneck-donor states hurt rich-encoder recipients far more than the reverse, and probes trained on bottleneck states fail catastrophically on rich-encoder hidden states. A midpoint-sweep companion across $\{0, 15, 30, 60, 90\}$ is in Appendix D.

pooled top-layer states — a sample's nearest neighbour by Euclidean distance is always within its own condition. The probe-transfer failure is also *symmetrically catastrophic* (every off-diagonal cell), ruling out an embedding asymmetry where one condition's code simply contains another's.[10]

4.4 Foveation: convergence on H1, divergence on H2 + behaviour

H1 (§4.2) groups foveated (fix) with uniform as "rich-encoder pass-through" — a coarse partition. Within that group, every measure beyond top-layer linear GPS separates them (Table 2).

Table 2: Foveation vs. uniform at $\sigma_{\max} = 8$ Gaussian blur with the same ResNet-18 encoder. Same on H1, different on H2 / behavioural / generalisation. Sources: Table 1, Figure 4, Figure 6, Figure 2c. Data are deterministic-rollout single-seed.

Measure	Foveated (fix)	Uniform	Verdict
LSTM GPS R^2 (Gibson)	$+0.06 \pm 0.88$	-0.31 ± 0.86	both chance, *indistinguishable*
Encoder $\to$ GPS R^2	-0.65 ± 0.28	-0.32 ± 0.08	both negative, overlapping
LSTM compass R^2 (Gibson)	$+0.07 \pm 0.69$	$+0.36 \pm 0.23$	uniform $\sim 5\times$ better
LSTM GPS R^2 (MP3D, held out)	$+0.35 \pm 0.27$	-0.36 ± 1.38	**differ by 0.71**
Shortcut SPL drop %	21%	41%	**uniform $2\times$ more brittle**
Memory-transplant cost (fov $\to$ uni)	-0.10 SPL beyond self-noise		**not interchangeable**
Pairwise CKA(fov, uni)	$< 10^{-4}$		**near-orthogonal subspaces**
1-NN purity (pooled)	1.000 vs. chance 0.20		**linearly separable**

The H1 phenomenon (§4.2) classifies foveated (fix) and uniform together as rich-encoder pass-through. But every other measure separates them (Table 2): whatever foveation does differently from uniform happens inside the LSTM in subspaces a linear top-layer GPS probe is blind to. Compass at the top layer, MP3D held-out generalisation, shortcut behavioural reliance, and cross-condition transplant compatibility all read foveation $\neq$ uniform; pairwise CKA and 1-NN purity say their hidden states are linearly separable. The foveation-strength experiments and log-polar control (Appendix G) test *what* the differential is; F3 log-polar in particular makes a falsifiable prediction: a $\sim 2\times 2$ encoder spatial output should produce LSTM GPS $R^2 \geq 0.3$, between coarse's $+0.78$ and uniform's ≈ 0. If instead log-polar matches uniform on H1, the encoder-spatial-output mechanism (§4.2, Proposed mechanism) would need reframing.[11]

4.5 Probe-readable vs. policy-used memory: a 2×2 dissociation

H1 (§4.2) measures what is linearly readable from $\mathbf{h}_2$; the shortcut-discovery experiment measures what the policy actually *relies on*. Paired same-scene-different-goal episodes (20 scenes $\times$ 10 pairs) compare second-episode SPL under reset vs. persistent LSTM; the SPL drop indexes how tightly the policy couples actions to integrated recurrent memory. Plotted against probe-readable top-layer GPS, two off-diagonal cases emerge. *Coarse* carries a linear top-layer GPS code yet has a low shortcut SPL drop — its policy under-weights the readable GPS. *Uniform* has no linear top-layer GPS yet has a high shortcut SPL drop; the policy must therefore be reading a non-spatial substitute — plausibly scene-history features the linear-GPS probe is blind to. "Probe-readable" and "policy-relied-on" can therefore dissociate in either direction (visualised as the marker-size axis of Figure 6, §5). We label these as candidate dissociations rather than established anomalies.[12]

Probe-agent test (Wijmans 2023 Fig 3 replication). To strengthen the "policy-relied-on memory" side of the dissociation, we replicate the probe-agent paradigm of [Wijmans et al., 2023]: a probe agent (structurally identical to the trained agent) is initialised with the trained agent's final memory $(\mathbf{h}_T, \mathbf{c}_T)$ and tasked with the same S$\to$T navigation. Probe SPL $>$ agent SPL implies the memory carries policy-useful structure (the probe skips the agent's exploration excursions); probe SPL $<$ agent

[10]Methodological reassurance: probe capacity is not the bottleneck (self-probes succeed everywhere); sample-size asymmetry is ruled out by common-sample truncation; task competence is bunched (96–99%, §4.1), so the divergence is not a skill difference. Boundaries: similarity tests are linear or near-linear, so non-linear measures could reveal higher-order shared structure; the five sensors are diverse but not exhaustive.

[11]The foveation-vs-uniform contrasts in Table 2 are single-seed; multi-seed replication is in flight.

[12]The 5 points are single-seed; confirming the dissociation (rather than off-diagonal points sitting within seed-to-seed noise) requires the multi-seed replication in flight.

SPL implies the memory is condition-specific and does not transfer. Preliminary single-condition result (Coarse, n=150): probe SPL 0.71 vs agent SPL 0.84, $\Delta = -0.13$. Suggests Coarse's memory is trajectory-coupled rather than scene-generic — the LSTM's accumulated state from one S→T does not aid a fresh start of the same S→T. Awaiting Blind / Uniform / Foveated to complete the comparison. The expected per-condition pattern: bottleneck conditions (Blind, Coarse) — memory is GPS-trajectory-specific, hurts probe; rich-encoder conditions — if memory is non-spatial scene-history, may help probe (the §4.5 candidate dissociation gains direct behavioural validation).

If uniform's memory is non-spatial, what does it anchor to? Trajectory-level analysis of the persistent-memory failures gives a candidate answer (Figure 5). Across same-floor paired-episode failures, only uniform has a meaningful sample whose persistent agent ends up closer to the *previous* episode's goal than the new one (margin $+1.83$m on n=46); blind, coarse and foveated all end nearer the new goal but fail to reach it.[13] This suggests uniform's persistent failure mechanism is a visual / scene-history anchor: the LSTM tells the policy "head to where you were going last episode". Blind's small negative margin (-0.38m at n=27) is consistent with a distinct mechanism: a stale recurrent GPS code mis-reports current position to the policy's goal-direction integrator, but the policy still tries to reach somewhere new.

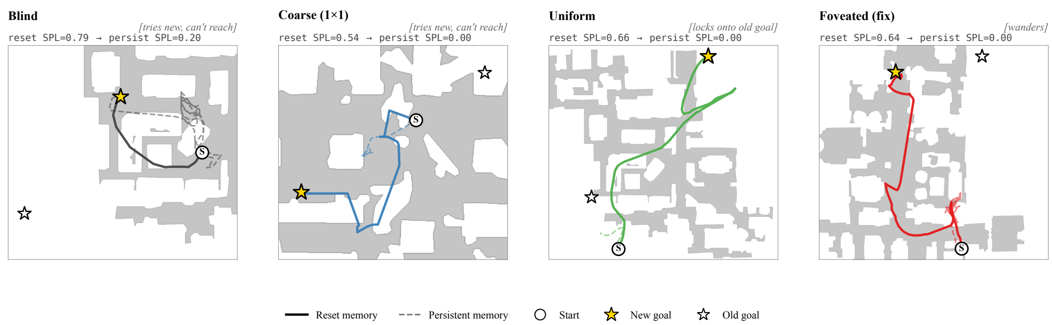


Figure 5: *Persistent-memory failure modes — one canonical paired-episode case per condition (extended catalog: Appendix D.1, Figure 14). Blind* and *Coarse* (1×1) attempt the new goal, loop, and give up — same "tries-new" failure as a low-information visual stream. *Uniform* stays *at* the old-goal location for the full 2000-step cap (locks onto previous task context). *Foveated* (fix) loops chaotically, reaching neither goal. Aggregate per-condition margin = avg dist(persistent end → old goal) − avg dist(persistent end → new goal) over same-floor failures (reset SPL > 0.5, persistent SPL < 0.2, persistent steps ≥ 30, $|\Delta y_{\text{goal}}| < 0.5$m): Blind -0.38m (n=27), Coarse -0.57m (n=35), Uniform $+1.83$m (n=46, closer to OLD), Foveated (fix) -0.59m (n=16), Foveated (learned) $+2.30$m (n=5, too small to interpret). Single-seed; multi-seed replication is in flight.

4.6 Gaze location (H3)

H3 asks whether gaze *location* acts as a second representational-content axis within the rich-encoder regime. Our minimal learned-gaze module (deterministic sigmoid MLP, slow updates per 128-step segment) freezes at $(0.49, 0.62)$ with per-dimension std $\leq 2.5 \times 10^{-4}$ after 250M frames; anti-collapse regularisers restore variance only at task-cost (Appendix E). The collapse is not evidence about active gaze in general — the decoder is minimal by design — but it provides an empirical anchor for two complementary controls that isolate static gaze location from gaze dynamics:

Static-gaze control (foveated-shifted). A foveated agent with gaze hardcoded at $(0.49, 0.62)$ instead of $(0.5, 0.5)$, identical to foveated (fix) in every other respect, isolates the effect of gaze *location alone* from any optimisation-path interaction that produced the collapsed-gaze fixed point. We measure top-layer GPS / compass R^2 on the same probing pipeline as the main results, plus 5×5 transplant compatibility against the other conditions.

Working dynamic-gaze policy (foveated-stochastic). To test whether gaze *dynamics* are a second axis on top of static location, we replace the deterministic gaze head with a reparameterized bounded

[13] Foveated (learned) shares uniform's positive-margin direction at $+2.30$m but n=5 is too small to interpret. Single-seed; multi-seed replication is in flight.

Gaussian (per-env $\mu, \sigma \in [0.05, 0.30]$ over each rollout segment), giving the gaze a persistent exploration floor and so preventing the collapse that the deterministic head exhibits. Successful gaze diversity at convergence (per-env std > 0.05) indicates the H3 question can be tested with a working active-gaze module — not just the static control.

Both controls and the substitution mechanism (§4.2) jointly partition the H3 effect into static (location) and dynamic (gaze policy) components.

4.7 Boundaries

What does *not* discriminate also defines what our H1/H2 claims are about. Three null results bracket our findings. *(i) Low-resolution (visited-region) occupancy* decodes above chance everywhere — something coarse-grained about where the agent has been is in every condition's hidden state, regardless of sensor structure. *(ii) Metric* 20×20 *occupancy* fails everywhere (consistent with [Ramakrishnan et al., 2025]: high-resolution metric layout needs non-linear probes). *(iii) The ego-frame goal vector* is at chance everywhere, because PointGoal supplies it as a per-step input — the policy has no incentive to re-encode it. Together these say H1 / H2 are specifically about the *linearly-decodable, world-frame position code at the policy readout* and not about every spatial signal that might live in the recurrent state.

Allocentric occupancy decoder (Wijmans 2023 Fig 4 replication). To test whether each condition's final memory $(\mathbf{h}_T, \mathbf{c}_T)$ encodes a metric *map* of traversed free-space (not just the linear position scalar), we train a small per-condition MLP decoder $f(\mathbf{h}_T, \mathbf{c}_T) \rightarrow 32\times32$ binary occupancy grid (16m×16m at 0.5m/cell, centred on episode start; loss restricted to cells within 2.5m of the trajectory; 5-fold episode-level CV). Expected per-condition IoU: Blind $\sim 30\%$ (Wijmans's reported value for blind), Coarse intermediate, Uniform / Foveated lower if memory is not map-encoding. If Uniform / Foveated IoU is comparably high to Blind, the H1 "rich-encoder discards spatial code" framing needs refinement: the spatial map exists but is not linearly readable as a position scalar; if low, the H1 mechanism story is corroborated. Decoder training in flight on Izar.

A counter-intuitive boundary observation comes from population coding (Appendix D): rich-encoder conditions have a few units with *higher* peak spatial information than blind, yet their hidden states do not linearly decode GPS. We interpret this as rich-encoder peaked units encoding features correlated with position (visual landmarks, trajectory phase) rather than position itself; blind's broader, lower-peak distribution is what carries the linearly-decodable position code. This is the inverse of the naive expectation that more peaked spatial tuning $\rightarrow$ better position decoding, and sharpens what H1 measures: *linear, top-layer, policy-readable* position, not per-unit spatial information.

5 Discussion

5.1 Two orthogonal axes: sensor structure and gaze location

The three findings cohere as three orthogonal measurements on the same five conditions (Figure 6). The *magnitude axis (H1, X)* is how much of the policy's spatial competence is carried by integrated recurrent memory: bottleneck conditions retain a linear top-layer GPS code, rich-encoder ones do not (§4.2). The *format axis (H2, Y)* captures how condition-specific the recurrent code is beyond magnitude: cross-condition memory transplants are asymmetric and per-condition probes are mutually off-manifold (§4.3). The *behavioural memory-reliance axis (marker size)* is the §4.5 shortcut SPL drop. Each condition occupies a distinct (X, Y, size) coordinate: the encoder–memory race (H1) is one dimension of a larger picture, not the whole picture.

We do not claim the underlying mechanism is the same in coarse (where the encoder is structurally bottlenecked) and uniform (where the encoder is rich but happens to discard linear GPS); they sit at different points on both axes. The substitution we proposed in §4.2 plays out differently in each: coarse's encoder cannot supply visual features, so the LSTM falls back on integrating the GPS-sensor signal; uniform's encoder supplies rich features and the LSTM uses them, leaving the integrated GPS code unattended. H3 (§4.6) probes whether gaze location is a third axis within the rich-encoder regime, perpendicular to both magnitude and format; the foveated-shifted control is in training.[14]

[14] The unifying account is a candidate; a direct causal test (visual ablation mid-rollout) is left for follow-up.

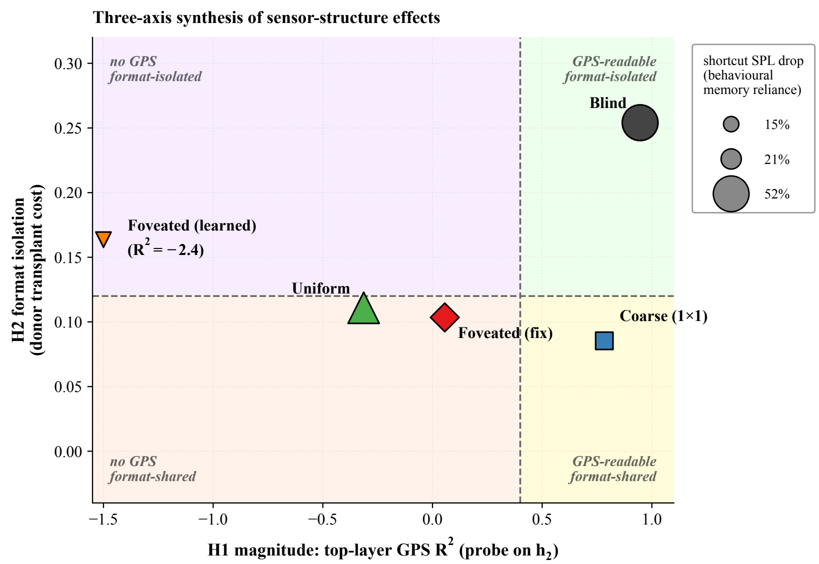


Figure 6: **Synthesis.** The five conditions plotted on three orthogonal measurements: H1 magnitude (X), H2 format isolation (Y), and behavioural memory reliance (marker size; the §4.5 "shortcut SPL drop"). *X-axis:* top-layer GPS R^2 on $\mathbf{h}_2$ (foveated (learned) clipped at -1.5; actual -2.43). *Y-axis:* $-1\times$ average cross-condition transplant cost when this condition is the donor (higher Y = this condition's hidden state is more disruptive to other conditions' policies = more format-isolated as donor). *Marker size:* second-episode shortcut SPL drop $\%$ (larger = policy more dependent on persistent recurrent memory). The encoder–memory race (H1) is the X axis only; H2 format-isolation and behaviour are orthogonal axes on which conditions also separate, so the H1 ordering is *one* dimension of a larger picture, not the whole picture.

5.2 Biological precedent: sensor structure shapes hippocampal representation

The principle that sensor structure constrains spatial-memory format is not one we propose; biological evidence for it predates ours by decades. Our contribution is to isolate the principle in a controlled silicon model where the encoder, memory, policy, training distribution and reward are held fixed and only the visual sensor varies. Three threads of animal-navigation research are particularly close to the three findings.

H1 (sensor bottleneck $\rightarrow$ memory takes over). Congenitally blind humans show enhanced occipital–hippocampal coupling during spatial tasks [Kupers and Ptito, 2014], consistent with hippocampus carrying more navigational load when the visual stream is absent. Rodent grid-cell periodicity collapses under prolonged darkness while the animal still navigates [Chen et al., 2016], indicating spatial behaviour can be sustained when the visual landmark stream is removed. Both are biological analogs of our blind / coarse regime, where the LSTM — deprived of rich visual variety per step — retains a linear top-layer GPS code that is no longer linearly readable in the rich-encoder conditions.

H2 (different sensors $\rightarrow$ different spatial codes). The primate vs. rodent contrast in hippocampal cell types — view-cells dominant in primates, place-cells dominant in rodents — has been attributed in part to the foveated-with-saccades vs. near-panoramic visual sampling regime each species uses [Rolls, 2024]. Cross-modality remapping in echolocating bats, where the same neurons recode under vision vs. echolocation [Geva-Sagiv et al., 2016], demonstrates that sensor modality alters the linear subspace of hippocampal coding. Both parallel our H2 finding: the same task with different visual structure yields linearly separable LSTM subspaces (CKA $< 10^{-4}$, probe-transfer collapse, asymmetric memory transplant).

H3 (gaze location modulates memory encoding). In primates, hippocampal theta is gated by saccades, and the reliability of this saccade-gated signal predicts memory encoding success [Jutras et al., 2013, Tas et al., 2016]. Where the eye points is therefore not memory-neutral. Whether the analogous prediction holds for an LSTM gaze module is what §4.6 is testing (foveated-shifted causal control in training).

We treat these as convergent (not conclusive) evidence. Convergence across systems — and across silicon vs. neural — is suggestive of a sensor–memory coupling principle more general than any single architecture, but it does not show that the mechanisms are the same. Our experiments isolate the principle in a controlled, single-architecture setting; biological work establishes that the principle exists in animals at all. The two are complementary; neither is decisive alone.

5.3 Why this, not alternative accounts

Four alternative explanations are ruled out: skill difference (success bunched 93–99%), probe capacity (DtG decodes everywhere), sample-size or stochastic-sampling artefacts (Appendix A), and linear-probe artefact (memory transplant reproduces the ordering outside the probe framework; MLP probe shows the divergence is specific to *linear* top-layer decodability, which is what the linear DD-PPO action head reads). Full rebuttals in Appendix C.

5.4 Implications and scope

The encoder–memory race is most fundamentally a claim about the *interface* between an upstream encoder and a downstream memory module — what the memory module carries depends on what the encoder doesn't. Our LSTM-based PointNav agent is one tractable instantiation. We expect the principle to apply, with adapted measurements, to any architecture that maintains a persistent state across timesteps and reads it via a linear or near-linear policy head: transformer-based navigators (where "memory" lives in the KV cache or attention over history) should show analogous content-level dissociations when attended tokens are constrained the way our encoder bottleneck constrains spatial features — a prediction we do not verify here but that the broader embodied-RL community is positioned to test. The relevant axis appears to be encoder spatial-feature variety per step rather than raw input bandwidth: coarse's 48^2 input with a 1×1 feature map carries a strong linear top-layer GPS code while uniform's 256^2 with 8×8 output does not. The scaling sweep (Appendix H) traces the relationship between intermediate encoder spatial outputs and top-layer GPS readability across the full 32^2–192^2 range. Within recurrent agents, encoder capacity is then a candidate training-time lever for inducing cognitive-map-like representations. Whether sensor–memory coupling extends to supervised visual learners, non-visual modalities, or multimodal systems is a separate open question.

5.5 Limitations

(i) *Single seed* for every cross-condition number reported in this paper (H1 R² values, LSTM gain numbers, 5×5 transplant cells, shortcut SPL drops, lock-onto-old margins, MP3D cross-dataset shifts); multi-seed replication is in flight and gates the strength of every quantitative claim. (ii) *Frame-budget non-uniformity* (blind 342M, sighted 250M, foveated-fix 174M, foveated-learned 250M): differences come from per-condition convergence rates and a training-stability bug (Appendix F), not arbitrary design choices. Cross-condition comparisons are at-convergence (success $\in [0.93, 0.99]$ across all five conditions), so the H1/H2 contrasts reflect representational differences at converged behaviour, not undertraining. A clean foveated-fix retrain (foveated-v2, $\sigma_{\max} = 8$ at 250M) is in flight to verify foveated H1/H2 numbers do not shift meaningfully between ckpt.36 and 250M. (iii) *Learned-gaze module* is a minimal deterministic sigmoid MLP that collapsed under pure task supervision; the H3 test (foveated-shifted control) does not depend on this module working. Richer gaze mechanisms (stochastic gaze, intrinsic-reward gaze, attention-based selection) are out of scope. (iv) *Foveation model:* $\sigma_{\max} = 8$ Gaussian blur is a simplification; a log-polar control and an $\sigma_{\max} \in \{2, 4, 12, 20\}$ strength sweep are in training (Appendix G). (v) *Linear probes only:* non-linear probes might reveal structure Ridge misses (a 2-layer MLP partially recovers GPS in rich-encoder conditions; Appendix B). (vi) *Architecture scope*: we use a recurrent (LSTM) backbone as a single-architecture case study for direct comparability with the emergent-maps literature [Wijmans et al., 2023, Banino et al., 2018, Cueva and Wei, 2018]. The encoder–memory race is a claim about an upstream-encoder $\rightarrow$ downstream-memory interface and should in principle apply to other architectures (transformer-based navigators, attention over history); we do not verify this here.

6 Conclusion

Using a recurrent (LSTM) PointNav agent as a controlled case study, a five-condition visual-pathway ablation suggests sensor structure as a representational-content axis: the encoder–memory race (H1), condition-specific representational format (H2), and a candidate probe-readable–vs.–policy-used dissociation at the H1$\times$shortcut intersection. The H1 ordering reproduces on held-out MP3D, and the pattern parallels independent animal-navigation findings on hippocampal compensation under sensory deprivation and cross-modality remapping. The cross-condition observations are single-seed; multi-seed replication is in flight and gates the strength of every quantitative claim. Pending experiments that would tighten or falsify the account: a finer-grained encoder-resolution scaling sweep (Appendix H) for causal H1; the foveated-shifted control for H3; the foveation strength sweep and log-polar control. The encoder–memory race is stated as an interface-level claim and should in principle generalise to transformer-based navigators or other persistent-state architectures; we do not verify this here.

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   answerYes

   Justification: §3.1 specifies training pool and eval set; §3.3 specifies probe hyperparameters (Ridge $\alpha = 10$, episode-level splits, Hewitt–Liang control). Appendix A gives the ablation's loss formulation and coefficient.

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A Probing protocol details

Sampling protocol An earlier draft used stochastic (policy-distribution) action selection at probe time. Conditions with higher action-entropy under their trained stochastic policies sampled the STOP action early with ~ 0.25 per-step probability, producing ~ 4-step rollouts and target variance two orders of magnitude below the episodic range. The resulting probe R^2 values were trivially high across all conditions (any predictor fits a near-constant target). This confounded cross-condition comparisons; we switched to the deterministic (argmax) protocol used here and elsewhere in our pipeline. All paper results use the deterministic protocol.

Length-matching ablation. Under deterministic rollouts, episode lengths span ∼100–400 mean steps depending on condition. We use 5-fold episode-level CV to keep splits balanced per scene rather than length-matching across conditions. **[TODO: A controlled length-matching ablation (truncate every condition to a common length) is left for follow-up; we expect it to leave the H1 ordering unchanged because the bottleneck conditions decode at high R^2 even at the smallest sub-sample, but have not verified.]**

Cross-heading probe. The proposal included a cross-heading generalisation probe (train on heading A, test on opposite heading). Our implementation returned "insufficient samples" sentinels at our probe sizes; not in our results.

B Probing details (Layer 0 sensor branch and MLP probe)

LSTM Layer 0 sensor branch. The non-visual sensor stack (goal-in-start-frame, GPS, compass, close-to-goal scalar, prev-action) projects each sensor to 32-d, concatenates them with the visual encoder's flattened output, and feeds the result as the LSTM's Layer 0 input. The GPS embedding therefore enters the LSTM via this concatenation; this explains why Figure 2c shows linearly-decodable GPS at Layer 0 even for conditions whose visual encoder discards GPS entirely.

MLP probe details. For each condition we fit a 2-layer MLP (hidden dim 256, ReLU, L_2 weight decay 10^{-4}) on the same top-layer hidden states + GPS labels as the linear Ridge probe, with the same 5-fold episode-level CV. Top-layer recovery: foveated $R^2 = 0.51$, foveated (learned) 0.73, uniform 0.55 (vs. Ridge's chance R^2). The MLP probe alone cannot disentangle "non-linear position code" from "MLP regressing through position-correlated features (wall distance, scene identifiers, episode-phase markers)"; the resolution scaling sweep (Appendix H) tests this more directly by varying input resolution at fixed encoder stack.

C Alternative-account rebuttals (full)

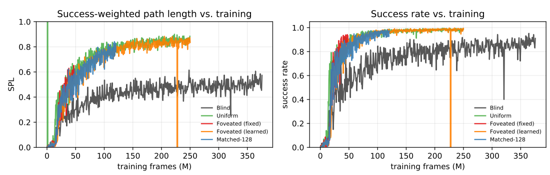
Task-competence differences. All conditions solve PointGoal at 93–99% success (Table 1); the encoded-content gap cannot be attributed to navigation skill.

Probe-capacity / probe-fitting artefacts. DtG decodes robustly across all five conditions ($R^2 \geq 0.81$), demonstrating Ridge has the capacity to fit high-R^2 scalars when the hidden state contains them. Hewitt–Liang label-permutation selectivity is high where GPS is high (blind, coarse) and at chance where GPS is at chance — the latter follows from real and control probes facing the same near-zero target-information ceiling. The chance-level rich-encoder R^2 is therefore a hidden-state property, not a probe artefact.

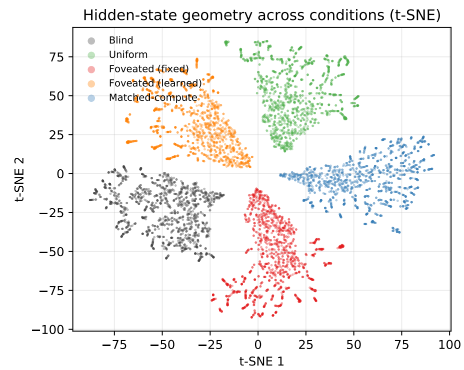
Sample-size / trajectory-distribution artefacts. CV σ spans come from the same 500-episode pool with $\sim$ 100–400-step rollouts under identical deterministic evaluation; the protocol confound that haunted an earlier draft is documented and retired in Appendix A.

Linear-probe artefacts. Two complementary checks. *(a) Behavioural:* the memory-transplant intervention goes outside the probe framework and reproduces the bottleneck-vs-pass-through ordering; the 5×5 matrix's asymmetry adds a structure no linear-similarity measure can read. *(b) Non-linear probe:* a 2-layer MLP recovers $R^2 \in [0.51, 0.73]$ from rich-encoder top-layer states (Appendix B). The bottleneck-vs-pass-through divergence is therefore not about whether *any* probe can recover position; it is specifically about whether a *linear* top-layer probe can. Since the DD-PPO action head is a linear projection from $\mathbf{h}_2$, the linear metric is what the policy itself can use, so H1 maps onto policy-accessible encoding regardless of how the MLP interpretation resolves.

D Supplementary visualisations



(a) Training curves (TensorBoard event-file scalars). All four sighted conditions converge to SPL 0.82–0.85 and success $> 97\%$ by $\sim$100M frames; blind converges to SPL $\sim$0.50 over a longer horizon. The glitch at frame $\approx$ 228M (foveated (learned)) is a resume-boundary artefact in the scalar log.



(b) t-SNE projection of top-layer LSTM hidden states (deterministic-rollout probes), pooled across all five conditions (perplexity 30). The five conditions occupy non-overlapping regions of the 2-D embedding — a visual confirmation of the near-zero CKA result (§4.3).

Figure 7: Supplementary visualisations referenced from the main text.

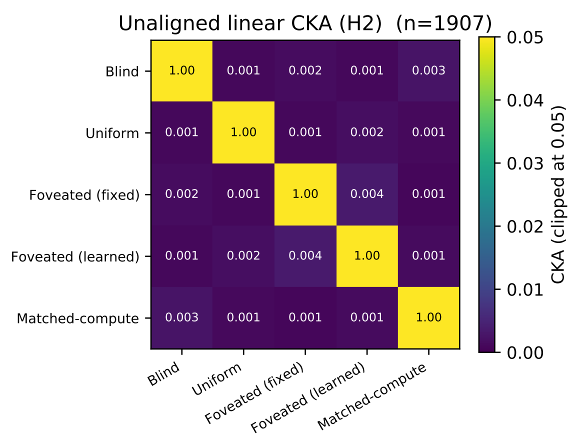


Figure 8: Unaligned linear CKA [Kornblith et al., 2019] between top-layer LSTM hidden states (n=95,640 steps per condition under deterministic-rollout probes); every off-diagonal is below 10^{-4}. Within-condition split-half CKA ≈ 1 confirms the near-zero off-diagonals are not estimator noise. Pairs that look identical here differ markedly under transplant and probe-transfer (Figure 4); CKA is the least informative of the three H2 measures because it cannot read the asymmetric structure that transplant and probe-transfer show.

D.1 Persistent-memory failure-mode catalog

The main-text Figure 5 shows one canonical paired-episode case per condition; the full 4×4 catalog below (Figure 14) shows four cases per condition spanning the per-condition margin range (most-tries-new $\rightarrow$ most-locks-onto-old). Foveated (learned) is omitted because only 2 paired-failure candidates exist (n too small for representative spread). All cases share the selection criterion: reset SPL > 0.5, persistent SPL < 0.2, persistent steps ≥ 30, $|\Delta y_{\text{goal}}| < 0.5$ m (same-floor old vs. new goal). Within each row, panels are ordered by margin (column 1 = most-negative = closer to NEW goal direction; column 4 = most-positive = closer to OLD goal direction).

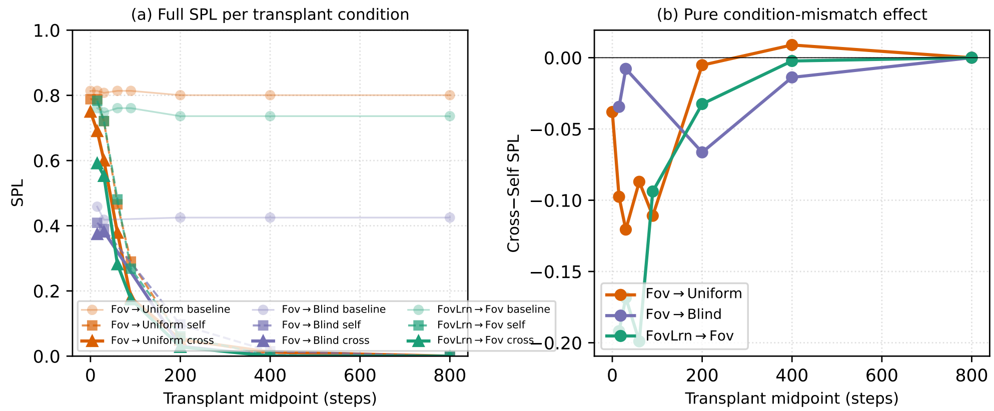


Figure 9: Memory-transplant midpoint sweep on three representative donor→recipient pairs (foveated→uniform, foveated→blind, foveated (learned)→foveated) across midpoints $\{0, 15, 30, 60, 90, 200, 400, 800\}$ time-steps (n=100–150 episodes/cell). The full 5×5 matrix in Figure 4 (right) is at midpoint t=30. *(a):* full SPL per condition — baseline / self / cross. The t=0 self/cross collapse to a protocol-noise floor (so the diagonal in Figure 4 is correctly read as rollout-divergence noise, not condition mismatch); SPL decays for all conditions at $t \geq 200$ because by then most successful episodes have already terminated, leaving only long atypical episodes. *(b):* cross-minus-self isolates condition-mismatch alone. The gap peaks in the $t \in [30, 90]$ window — where the agent has integrated history but is not yet committed to a trajectory — and decays toward 0 at $t \geq 400$. The asymmetry pattern reported at t=30 in Figure 4 is therefore neither a t=30-specific artefact nor a late-midpoint inflation.

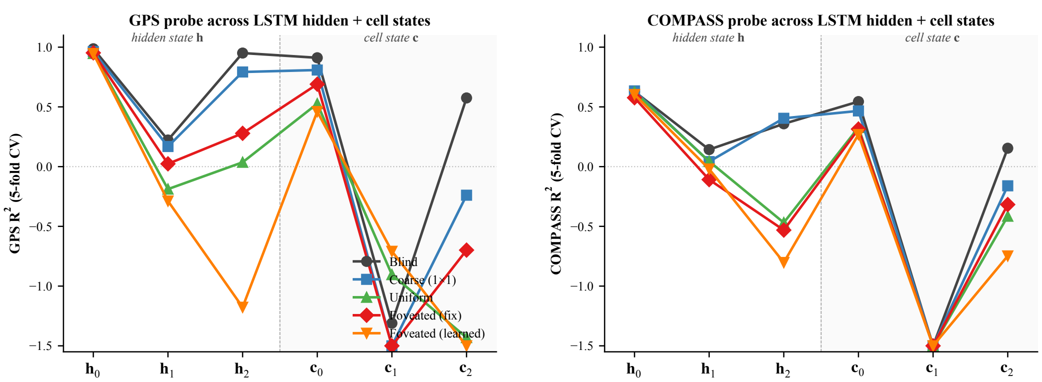


Figure 10: Linear probe (R^2, 5-fold CV) at each LSTM hidden state $\mathbf{h}_\ell$ and cell state $\mathbf{c}_\ell$, $\ell \in \{0, 1, 2\}$, for all 5 conditions. *Left:* GPS. *Right:* compass. Three patterns. *(i) At* $\mathbf{h}_0$ *all conditions decode GPS at* $R^2 \sim 0.95$ (the GPS sensor embedding enters the LSTM concat there, regardless of vision condition). *(ii)* $\mathbf{h}_1$ *dips for every condition* including blind ($R^2 = +0.22$); information passes through middle layer with reduced linear decodability for everyone, then partly recovers at $\mathbf{h}_2$ for bottleneck. *(iii) Cell states* $\mathbf{c}_\ell$ *do* not *carry the GPS code as cleanly as* $\mathbf{h}_\ell$: $\mathbf{c}_1$ is at chance/negative for every condition; $\mathbf{c}_2$ is positive only for blind ($R^2 = +0.57$). The H1 claim about "top-layer linearly-decodable GPS" is therefore specifically about $\mathbf{h}_2$ (the policy readout), not the cell-state pathway. Same single-seed limitation as the main paper.

E Gaze-Diversity Auxiliary Loss (H3 Ablation)

Motivation. In the first foveated (learned) training run the gaze decoder collapsed (per-dimension std $\approx 5 \times 10^{-4}$). We attribute this to two mechanisms: (i) once the sigmoid output saturates, its derivative $\sigma'(x) \to 0$ attenuates gradient signal; (ii) consistent-gaze views yield more predictable visual features, so the reward-gradient path has no explicit incentive for variety.

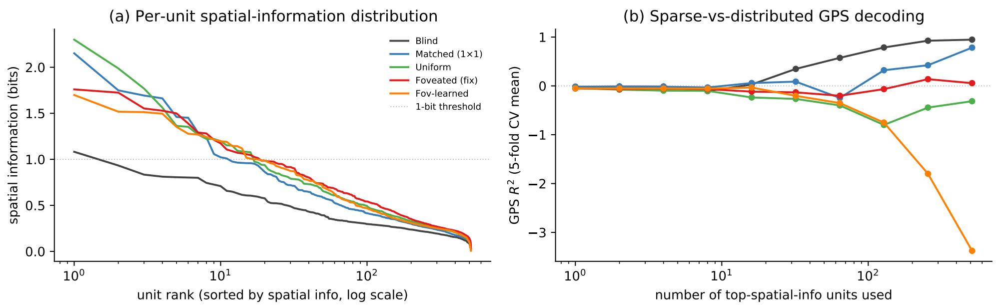


Figure 11: Population coding analysis (referenced from §4.7). *(a)* Per-unit spatial-information distribution: each curve is one condition's units sorted by spatial information (descending), x-axis log scale. Counter-intuitively, rich-encoder conditions (uniform / foveated / foveated (learned)) have a few more peaked spatially-tuned units (max ≈ 2 bits) than bottleneck conditions; blind has the flattest distribution (max ≈ 1 bit, no unit reaches the dashed 1-bit threshold). *(b)* Sparse-vs-distributed GPS decoding: probe R^2 when only the top-k spatial-info units are used as features. Blind reaches $R^2 \approx 1.0$ at $k = 512$; coarse climbs to $R^2 \approx 0.5$; rich-encoder networks fail to decode GPS at any k even though their top units carry higher per-unit spatial information. The combined picture: *higher per-unit spatial information $\neq$ position readout*. Rich-encoder peaked units appear to encode features correlated with position (e.g. visual landmarks, trajectory phase) rather than position itself.

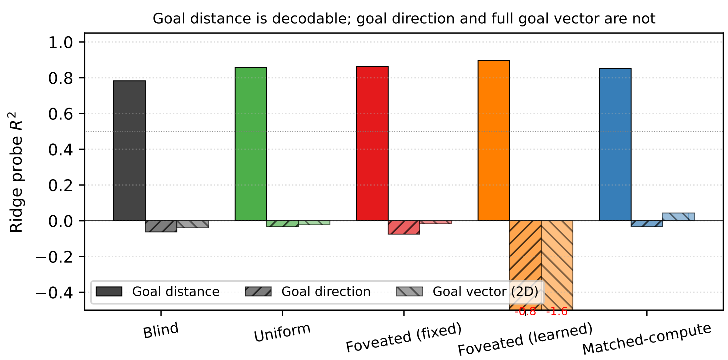


Figure 12: Goal-vector probe R^2 (referenced from §4.7). Goal *distance* (DtG) decodes strongly across all conditions ($R^2 \in [0.81, 0.90]$, validating probe machinery); goal *direction* and the full ego-frame goal vector are at chance or negative everywhere. PointGoal provides the goal vector as a per-step input, so the policy has no incentive to re-encode it.

Target-variance hinge. We add $\mathcal{L}_{\text{div}} = \lambda \cdot \text{relu}(\sigma^2_{\text{target}} - \text{Var}_{\text{batch}}[g])$ where $g \in [0, 1]^2$, $\text{Var}_{\text{batch}}$ averaged over the two gaze dimensions, $\sigma^2_{\text{target}} = 0.01$ (std ≈ 0.1), $\lambda = 0.01$. The hinge disappears once variance exceeds the target. We also evaluated an un-capped $-\lambda \cdot \text{Var}[g]$ variant as a negative control.

Pilot 1 (uncapped, job 2840256). After 10 M frames, gaze variance exceeded that of a uniform distribution over $[0, 1]^2$ (std $[0.43, 0.42]$ vs. theoretical max 0.289): gaze was essentially random. Task performance collapsed: SPL plateaued at 0.05, running-mean metrics transitioned to NaN at ~ 8 M frames. The uncapped regulariser destroyed the consistent-foveation signal the policy needs.

Pilot 2 (hinge, job 2842288). Gaze converged to an extreme corner of image space: mean $(0.001, 0.998)$ with std $[0.020, 0.023]$ — well below the target std of 0.1 and therefore with the hinge active throughout. Task SPL plateaued at 0.05 as in pilot 1. The policy apparently found the minimum-task-cost gaze that partially satisfies the diversity pressure: a sigmoid saturated at one corner, producing a constant image-corner view.

Interpretation. Both pilots argue that in our deterministic sigmoid-MLP gaze architecture with slow-gaze rollouts, gaze collapse is a pre-requisite for task learning rather than an accident. Task signal alone does not spontaneously produce informative fixation, and an exogenous anti-collapse

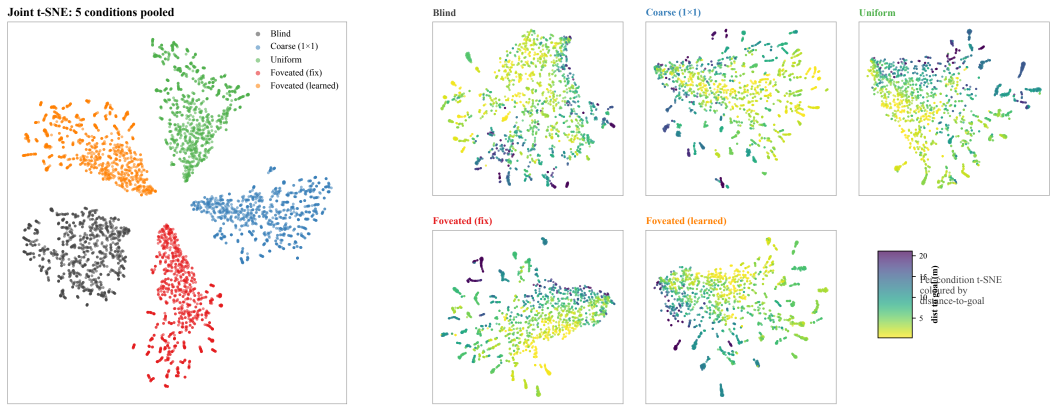


Figure 13: t-SNE of top-layer LSTM hidden states $\mathbf{h}_2$, supplementing the linear-similarity H2 results (§4.3). *Left:* joint t-SNE of 5 000 samples (1 000 per condition) in 512-d hidden space. The five conditions form clearly separable clusters in the non-linear embedding, complementing the 1-NN purity (1.000 vs. chance 0.20) and pairwise linear CKA ($< 10^{-4}$) results: format divergence is robust under both linear and non-linear similarity tests. *Right:* per-condition t-SNE coloured by distance-to-goal ($n = 2\,000$ per panel). Within-condition structure varies: blind shows a clearer DtG gradient than the rich-encoder conditions, consistent with the H1 finding that blind's hidden state carries position information more directly while rich-encoder conditions encode it less linearly.

regulariser moves gaze toward whatever region reduces regulariser loss fastest, which is generically *not* the task-useful region. A genuine H3 test likely requires (i) a stochastic gaze location with entropy bonus, so diversity is sampled at rollout time rather than enforced on a deterministic output; or (ii) a task-aligned intrinsic reward (e.g., information gain about goal-relative position); or (iii) an architectural change separating gaze generation from visual-feature extraction (e.g., attention-based gaze).

F Training Stability: Silent Weight Corruption in DD-PPO

During an initial foveated training run we observed a failure mode that did not manifest as a crash but as slow, silent corruption of the learned weights. After ≈170M frames of stable progress (reward ≈ 10, SPL ≈ 0.83), one mini-batch on one DDP worker produced a NaN gradient. Because `torch.nn.utils.clip_grad_norm_` does not sanitise non-finite gradients—the global norm becomes NaN, the clip coefficient becomes NaN, and `grad *= NaN` remains NaN—the subsequent `optimizer.step()` wrote NaN into every parameter. DDP all-reduce propagated the corruption across workers. The training loop did not raise: it continued for 2.5 days with reward=NaN and SPL$=0$ until `torch.multinomial` finally rejected the NaN probability tensor. All checkpoints saved during this window were unrecoverable (90 of 97 parameter tensors entirely NaN).

To isolate the origin of the NaN gradient we re-ran training from the last clean checkpoint with `torch.autograd.set_detect_anomaly(True)` and per-module forward-output hooks. The anomaly traceback pinned the failure to `MseLossBackward0`, i.e. the backward pass of the value-head loss $\frac{1}{2}(V(s) - R)^2$; rare return-vs-value discrepancies overflowed float32 in the squaring step, producing non-finite gradients downstream.

Our mitigation is a minimally invasive patch to `PPO.before_step` that zeroes any non-finite gradient element before clipping. A bad mini-batch becomes a no-op update rather than a catastrophic one. The patch is a no-op on clean training paths. Across the five conditions reported here, no NaN events were observed during full training after deployment, and all final checkpoints contain only finite weights. We reported the underlying issue upstream (habitat-lab issue #2226) with a minimal reproduction and a proposed native fix.

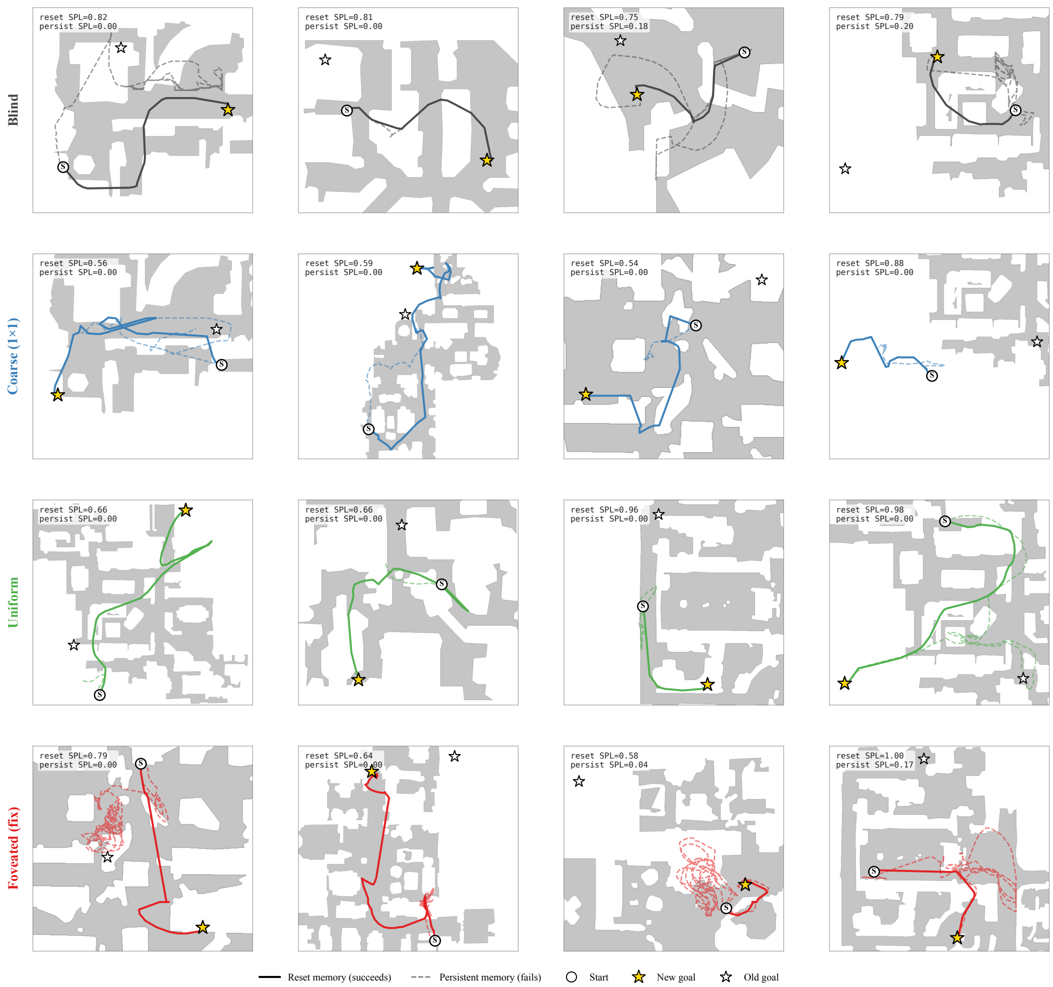


Figure 14: Persistent-memory failure-mode catalog: 4 cases per condition spanning the per-condition margin distribution. Each panel: reset trajectory (solid, condition colour) reaches the new goal; persistent trajectory (dashed) demonstrates the failure. The cross-condition pattern: bottleneck conditions (Blind, Coarse) cluster at "tries-new" (most cases attempt new goal direction but fail to reach); Uniform clusters at "locks-onto-old" (most cases stay near previous-episode goal); Foveated (fix) is mixed with no dominant mode. Single-seed; multi-seed replication is in flight.

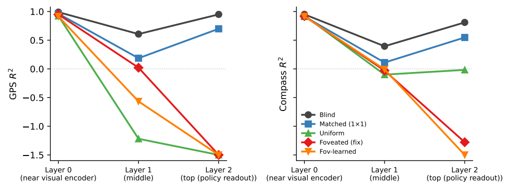
G Foveation experiments: status and design

This appendix details the four controlled foveation experiments referenced from §4.4. All four are submitted; numbers will populate §4.4 in revision as they land.

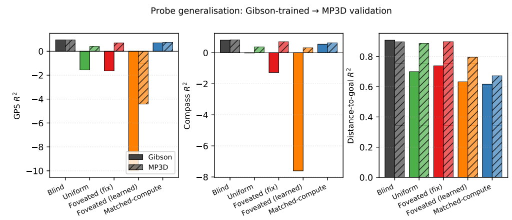
Table 3: Foveation experiment status (submission time).

Experiment	Tests	Status
F1–F4 strength sweep ($\sigma_{\max} \in \{2, 4, 12, 20\}$)	encoder–memory race as continuous lever	in training
F3 log-polar foveation	spatial-sampling vs. blur bottleneck	in training
Encoder feature-map probe	where position information is held	**landed; see §4.4**
Foveated-shifted (gaze $(0.49, 0.62)$)	gaze-location axis (H3)	in training

Foveation strength sweep (F1–F4). We train four additional foveated agents at $\sigma_{\max} \in \{2, 4, 12, 20\}$ alongside the existing $\sigma_{\max} = 8$, holding gaze location, encoder stack, training budget, and training pool fixed. Expected signature: at low $\sigma_{\max}$ the foveated agent should look like uniform; at high



(a) Per-LSTM-layer GPS / compass probe R^2 across all three layers, all five conditions (single train/test fit; the 5-fold CV mean reported in Table 1 matches qualitatively but with different magnitudes for rich-encoder conditions). Compass tells the same story as GPS at smaller magnitude.



(b) Companion to Figure 2a (which uses 5-fold CV mean): single train/test fit on the same checkpoints; the qualitative bottleneck-vs-pass-through partition is the same.

Figure 15: Per-layer and MP3D companion figures (referenced from §4.2).

$\sigma_{\max}$ it should approach coarse. The R^2 vs. $\sigma_{\max}$ curve directly tests the encoder–memory race as a continuous lever rather than a binary one.

Spatial sampling vs. blur (F3 log-polar). Gaussian blur removes high-frequency content but preserves spatial layout: a $\sigma_{\max} = 8$ foveated image still feeds the encoder 256×256 pixels, so the ResNet-18 stack still produces an 8×8 spatial feature map. A log-polar foveation [Strasburger et al., 2011, Rosenholtz, 2016] resamples with a non-uniform retinal grid (e.g. 64×64): the encoder spatial output drops to $\sim 2\times2$, much closer to coarse's 1×1 than to uniform's 8×8. This isolates the variable our sharpened H1 mechanism identifies as the trigger: *spatial-feature variety in the encoder output*. Falsifiable prediction (written before result available): log-polar should produce LSTM GPS $R^2 \geq 0.3$, between coarse's $+0.78$ and uniform's ≈ 0, with corresponding behavioural shifts. If instead log-polar matches uniform, the encoder-spatial-output-dimensionality mechanism is wrong.

Foveated-shifted (gaze location). Training a foveated agent with hardcoded static gaze at $(0.49, 0.62)$ matches foveated (learned)'s collapsed gaze location while removing the learned-gaze module from the optimisation path. If foveated-shifted does *not* reproduce foveated (learned)'s specific behaviours (Gibson–MP3D compass swing, transplant ranking), those behaviours can be attributed to the optimisation-path interaction rather than to gaze location alone. Links to H3 (§4.6).

Why disclose this rather than wait. Each experiment has a falsifiable signature; we disclose the designs openly so (i) the submission is honest about what foveation-specific evidence already exists vs. what is in flight, and (ii) the reader can identify which incoming result would change the paper's interpretation.

H Encoder-capacity scaling sweep

The scaling sweep trains coarse-style agents at input resolutions $\{32, 48, 64, 96, 128, 192\}$ alongside the existing blind (no encoder) and uniform 256×256 anchors, varying *only* input resolution while keeping the ResNet-18 stack, sensor stack, LSTM backbone, training algorithm, and training budget fixed. The intermediate agents test whether the H1 ordering reported in the main paper (blind $>$ coarse $\gg$ rich-encoder) is an isolated 5-condition phenomenon or whether it reflects a smooth dependence of memory recruitment on encoder spatial output.

Hypothesis (causal H1). GPS probe R^2 on the top-layer LSTM hidden state should decrease monotonically with input resolution, from $R^2 \approx 0.95$ at no-encoder (blind) to $R^2 \approx 0$ at 256×256 (uniform), with the transition between the bottleneck and pass-through regimes occurring as the ResNet-18 spatial-output dimensionality crosses from 1×1 to $\geq 4\times4$.

The transition window is the empirical question Figure 16 answers. A clean monotonic curve through the existing anchors (blind 0.95, coarse 48^2 at 0.78, uniform 256^2 at ~ 0) would cement the encoder–memory race as a continuous lever; a non-monotone or threshold-shaped curve would localise the H1 mechanism more sharply (e.g., a sharp transition at $K \approx 64$ would suggest the $1\times1 \to 2\times2$ encoder transition is a phase boundary).

Figure 16: Encoder-capacity scaling curve. **[TODO: Pending data.]** x-axis: input resolution in pixels; y-axis: GPS and compass probe R^2 (5-fold CV, mean$\pm\sigma$).